

Breast Ultrasound Lesion Segmentation: Research Write-up

Version 1.0.0 · Site-only research report (not a journal submission)

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Live demo: <https://surabhif.github.io/Breast-ultrasound-lesion-segmentation/>

Repository: <https://github.com/surabhif/Breast-ultrasound-lesion-segmentation>

Software DOI: <https://doi.org/10.5281/zenodo.23286597>

Research demo — not for clinical use. Not for diagnosis, screening, or care decisions.

1. Abstract

Breast ultrasound helps doctors look at breast lumps, especially when mammograms are hard to read in dense tissue. This student project builds a program that outlines those lumps on ultrasound pictures and estimates how well the program tells benign from cancerous lumps. It was trained on one public image collection and checked on three others from different hospitals. The main quality measure is an outline-overlap score (Dice), which is high when the computer outline matches the expert outline. On held-out training-collection images, the browser version reaches 0.697 overall and 0.764 on images that contain a lump. On two outside collections, outline scores stay similar (0.714 and 0.627), while telling benign from cancerous gets harder after the hospital and scanner change. On a third outside collection, empty or normal images remain difficult: false outlines pull the all-image score down to 0.386 even though the lesion-only score is 0.679. The write-up also studies near-duplicate frames, burned-in measurement marks, and whether outlines stay stable when a picture is flipped. A later mixed-collection retrain improved some outside scores but did not pass a pre-agreed replacement rule, so the site still serves the original model. This is an educational research demonstration, not a medical device, and must not be used for diagnosis or care decisions.

Key findings in plain English

- On held-out training-collection images, the browser model's outline-overlap score is 0.697 overall and 0.764 on images that actually contain a lump.
- On outside hospitals, outline quality stays in a similar ballpark on BUS-BRA (0.714) and BrEaST (0.627), but how well the program tells benign from cancerous drops (0.638 / 0.721 vs 0.931 inside).
- On BUS-UCLM, normal images without lumps drive many false outlines (320/413); the lesion-only outline-overlap score 0.679 is the fairer comparison.
- Near-duplicate frames are common (322 images in multi-member groups). A short leakage check did not show random splits inflating validation scores ($\Delta \approx -0.022$), but patient IDs are still missing.
- Erasing measurement marks and a later multi-collection retrain improved some outside scores but did not replace the published model under the pre-agreed swap rule (clean-subset score moved by about -0.009).
- This is an educational research demo only — not for diagnosis, screening, or care decisions.

2. Background: why outlining lumps on ultrasound matters

Breast ultrasound uses high-frequency sound waves to form a live picture of breast tissue. Clinicians often add ultrasound after a mammogram or a physical exam when they need a closer look at a mass, especially in dense breasts where X-ray contrast is limited (Mendelson et al., ACR BI-RADS Ultrasound, 2013). For a patient, the practical questions are simple: Is there a lump? How large is it? Does it look more concerning or more reassuring? Those answers influence biopsy decisions and, later, surgical planning.

Outlining a lump — drawing its boundary on the image — is the spatial version of that conversation. A careful outline supports size estimates, shape descriptors used in Breast Imaging Reporting and Data System (BI-RADS) language, and communication among radiologists, surgeons, and patients. Automated outlining is attractive for education and research because public datasets now include expert masks. It is also easy to over-claim: a computer outline is not a surgical margin, an imaging diameter is not pathologic tumor stage, and a model score is not a BI-RADS assessment category (Moran et al., SSO-ASTRO margin guideline, 2014).

This project asks a narrower, honest question suitable for a student portfolio: can a small model outline lesions in the browser, and do the published numbers survive outside the training hospital's pictures? The site pairs the demo with BI-RADS context and a surgeon's-view page so readers see what the numbers do not mean.

From a patient's point of view, an outline is a picture of extent: how much of the frame looks abnormal, and whether the edge looks smooth or irregular. Surgeons later care about margins in tissue, not pixels, which is why this write-up repeatedly separates imaging size from pathologic stage. From a student's point of view, the same outline is a measurable prediction that can be scored, stress-tested on outside hospitals, and shipped in a browser so anyone can see the failure modes without installing a machine-learning stack.

3. Related work

U-Net-style encoder-decoder networks remain a default for biomedical segmentation (Ronneberger et al., 2015). On the Breast Ultrasound Images (BUSI) dataset, recent papers report a wide range of scores depending on whether splits are random, whether near-duplicates are removed, and whether normal (no-lesion) images are included. UNeXt (Valanarasu & Patel, MICCAI 2022) reports F1 about 0.79 and intersection-over-union (IoU) about 0.67 on an 80–20 random split of benign and malignant BUSI frames at 256×256 — a useful context number, but not an apples-to-apples match for this project's grouped split and per-image outline-overlap score. Musah et al. (2025) show that BUSI scores fall after de-duplication and fall further when a BUSI-trained model is tested on BrEaST (reported Dice about 0.49 in their setting). Pawłowska et al. (2023) document roughly 235 duplicated BUSI images (~19%), which is why this project groups perceptual near-duplicates before splitting.

This work's served browser model reaches outline-overlap score 0.697 (lesion-only 0.764) on the grouped held-out BUSI test, and external outline-overlap scores 0.714 (BUS-BRA), 0.627 (BrEaST), and lesion-only 0.679 (BUS-UCLM). Classification area under the curve drops from 0.931 internally to 0.638 / 0.721 / 0.780 externally — the same direction Wang (2026) reports for classification under dataset shift. Published tables are context, not a leaderboard; protocol differences dominate small score gaps. Full row-by-row notes live in the project's literature-comparison document.

Positioning matters as much as the table. This is not a claim to beat UNeXt or nnU-Net on a public leaderboard. It is a claim that a single frozen browser artifact has internal scores, external scores, cleaning experiments, and a written rule for when the artifact may change — and that those pieces are visible next to the demo.

Paper / setting	Split & notes	Reported figure	How it compares here
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This work v1.0.0 served	BUSI grouped held-out n=112	Outline-overlap 0.697; lesion 0.764; AUC 0.931	Reference row
This work external	Frozen weights; no retune	BUS-BRA 0.714; BrEaST 0.627; BUS-UCLM lesion 0.679	Transfer check
Valanarasu & Patel, UNeXt (2022)	BUSI B+M; 80-20 random; 256 ²	F1 $\approx$ 0.79; IoU $\approx$ 0.67	Partly comparable (random split; different aggregation)
Musah et al. (2025)	BUSI→BrEaST OOD among other settings	BUSI→BrEaST Dice $\approx$ 0.49 in their setting	Same OOD direction; different model/resolution
Pawłowska et al. (2023)	BUSI audit	~235 duplicates (~19%)	Motivation for near-duplicate grouping

Takeaway: published BUSI scores are easy to misread unless the split and metric definition match; this project optimizes for honest transfer reporting, not a leaderboard win.

4. Data

Four public breast-ultrasound collections are used. BUSI is the training and internal test source. BUS-BRA, BrEaST, and BUS-UCLM are external checks of the frozen served model with no threshold retuning. BUSI provides no patient identifiers; the other three do. Annotation marks (calipers, text overlays) were flagged heuristically on 46.9% of BUSI frames (366 / 780). Perceptual near-duplicate grouping produced 601 groups, with 322 images in multi-member groups.

Dataset	Images	Patients	Source country	Scanner notes	License	Excluded / notes
BUSI (Al-Dhabyani et al., 2020)	780	Not published	Egypt (Baheya / Cairo University collection)	Clinical ultrasound archive; device mix not standardized in the release	Citation required; full-archive redistribution rights unclear — not redistributed here	Multi-mask lesions OR-merged; near-duplicate groups kept intact across splits
BUS-BRA (Gómez-Flores et al., 2024)	1875	1064	Brazil	GE Logiq 5/7, Toshiba Aplio 300, U-Systems (device field in release)	Creative Commons Attribution 4.0 (Zenodo)	No normal class in this release; used as external test of frozen weights
BrEaST (Pawłowska et al., 2024)	256	256	Poland	Clinical scanners with physical pixel size in metadata	Creative Commons Attribution 4.0 (TCIA)	Freehand tumour masks; few normals; used as external test
BUS-UCLM (Vallez et al., 2025)	640	38	Spain (Castilla-La Mancha)	Siemens ACUSON S2000	Creative Commons Attribution 4.0 (Mendeley)	43 Doppler/combined frames excluded; images scored locally, not redistributed

Takeaway: four public collections from different countries and scanners; only BUSI trains the published model, and BUS-UCLM drops Doppler/combined frames before scoring.

Sources: dataset papers cited above; counts from committed results JSON (cleaning_experiment.json; external busbra/breast/busuclm.json).

BUS-UCLM (Vallez et al., 2025; Mendeley Creative Commons Attribution 4.0) is scored for the frozen served browser model: $n=640$ / 38 patients (43 Doppler/combined frames excluded). All-image outline-overlap score 0.386 [0.326, 0.449]; lesion outline-overlap score 0.679 [0.593, 0.755]; area under the receiver-operating curve (AUC) 0.780. Normal false positives dominate the all-image figure (320/413) — the same known weakness as on BUSI (12/19). Prefer the lesion outline-overlap score for cross-dataset comparison (between BrEaST and BUS-BRA).

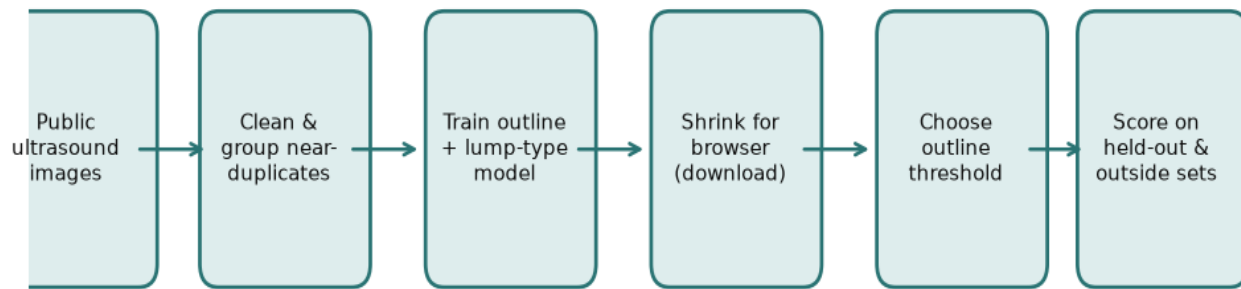
Internal split sizes after grouped stratification: train 534, validation 134, held-out test 112.

5. Methods

5.1 Pipeline overview

Figure 1 summarizes the path from public images to browser scores. Every served number on the site is produced by regenerating metrics from committed results files so the demo and the write-up cannot silently diverge.

Figure 1. End-to-end pipeline (plain overview)



Takeaway: images are cleaned and grouped, a compact model is trained, shrunk for browser download, thresholded, then scored on held-out and outside sets.

5.2 Preprocessing

Each image is loaded as red-green-blue, resized to 160×160 with a shared half-pixel-center bilinear resize (identical in Python evaluation and in the browser), scaled to [0, 1], and normalized with ImageNet channel means and standard deviations. Expert masks are resized with nearest-neighbour interpolation and binarized; when a lesion has several mask files they are combined with a logical OR so any marked region counts.

5.3 Near-duplicate grouping

Because BUSI does not publish patient identifiers, near-identical frames can otherwise appear in both training and testing. Frames are grouped with a perceptual hash so that an entire near-duplicate group stays on one side of each split. That does not prove zero leakage — only that the obvious near-copies are not split apart.

5.4 Model, explained intuitively

The network is a compact encoder-decoder in the U-Net family (Ronneberger et al., 2015) with a ResNet-18 ImageNet encoder (He et al., 2016). Intuitively, the encoder compresses the ultrasound into a smaller grid of features; the decoder expands those features back to a full-size outline map. A second head reads the same compressed features and outputs a single score for benign versus malignant. Training uses paired image-mask augmentations, stratified grouped splits, and checkpoint selection on lesion outline-overlap score on the validation set.

5.5 Shrinking the model for the browser

The full-precision training checkpoint is about 46.3 megabytes. For the website it is exported to the Open Neural Network Exchange (ONNX) format and dynamically quantized to 8-bit integer weights (often called INT8), yielding a 11.7 MB download. The browser runs that artifact with onnxruntime-web (WebAssembly) inside a Web Worker so the page stays responsive. Post-processing matches Python evaluation: probability threshold 0.4 and minimum connected-component area 40 pixels.

5.6 Threshold choice

The outline threshold 0.4 and minimum area 40 were chosen on BUSI validation lesion outline-overlap score and then frozen. External sets reuse the same operating point with no retuning — an intentional honesty choice that can look worse than a tuned baseline but answers the transfer question readers actually care about.

5.7 Training details in brief

Training uses the grouped BUSI split (train 534, validation 134, test 112). The optimization target emphasizes overlap with expert masks; the classification head is trained jointly so the same backbone supports both outlining and the benign-versus-malignant score. Checkpoints are picked by validation lesion outline-overlap score, not by hunting for the best held-out test number after the fact. The served fingerprint (sha256 prefix 0bbf529d...) is the artifact evaluated everywhere in this write-up unless a section explicitly names a v2 experiment candidate.

Evaluation code paths are shared: the same resize, normalization, threshold, and minimum-area filter run in Python metrics and in the browser worker. That parity is why the Results page can honestly say the site numbers are the served-model numbers, not a prettier training checkpoint left behind in a notebook.

6. Results

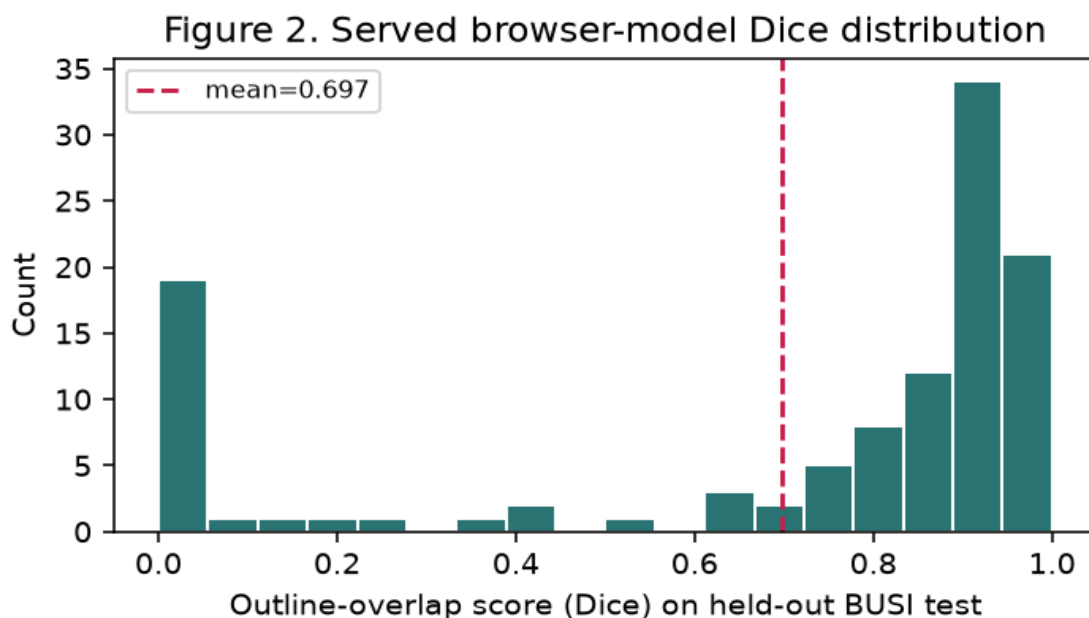
6.1 Internal results (held-out BUSI test)

Table 2 compares the full-precision training checkpoint with the served browser model on the same grouped test split (n=112). Overall outline-overlap scores are close (0.686 vs 0.697); lesion-only scores are 0.751 vs 0.764. Per-class served outline-overlap scores: benign 0.749 (n=63), malignant 0.797 (n=30). Normal images remain hard: non-empty predicted masks on 12/19 normals.

Metric	Full-precision training	Served browser model
Outline-overlap score (all)	0.686	0.697
Outline-overlap score (lesion-only)	0.751	0.764
Intersection-over-union (IoU)	0.618	0.630
AUC (benign vs malignant)	0.939	0.931
Sensitivity / Specificity	0.867 / 0.873	0.867 / 0.841
Expected calibration error (ECE)	0.111	0.111
Normal false-positive masks	12/19	12/19

Takeaway: shrinking the model for the browser barely changes outline quality on the held-out BUSI test; empty/normal images are still the weak spot.

Sources: results/full_run.json, results/served_int8_test.json.



Takeaway: most held-out BUSI test cases land at high outline-overlap scores, with a long left tail of hard misses and normal false positives.

6.2 External validation

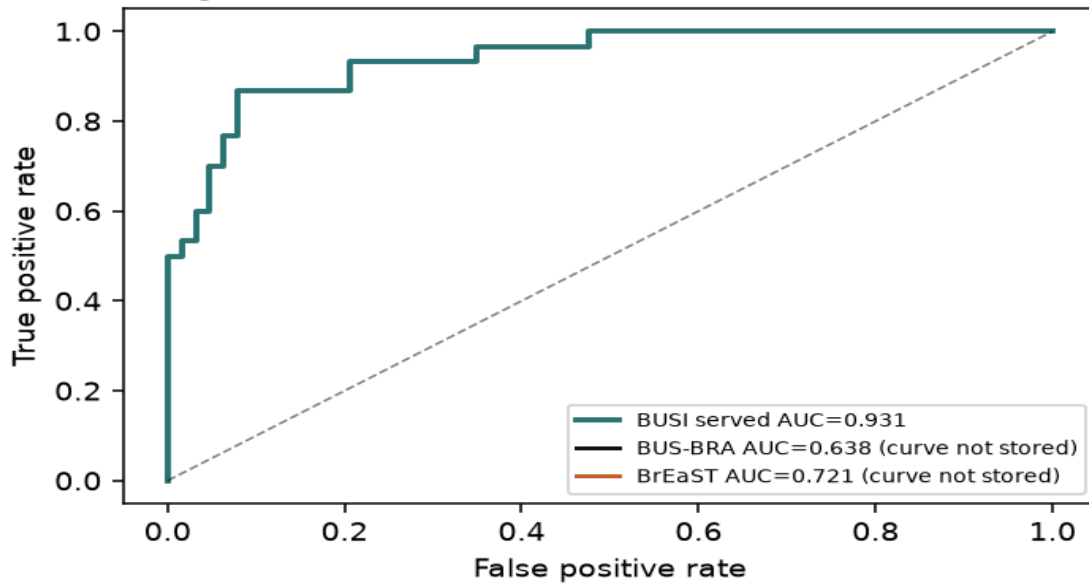
Frozen served weights were scored without tuning on BUS-BRA (n=1875, 1064 patients), BrEaST (n=256, 256 patients), and BUS-UCLM (n=640, 38 patients). Per-class outline-overlap scores: BUS-BRA benign 0.710 / malignant 0.721; BrEaST benign 0.593 / malignant 0.685; BUS-UCLM benign 0.702 / malignant 0.628.

Set	Outline-overlap	Lesion-only	AUC
BUSI internal (served)	0.697	0.764	0.931
BUS-BRA	0.714	0.714	0.638
BrEaST	0.627	0.629	0.721
BUS-UCLM	0.386	0.679	0.780

Takeaway: outlines transfer better than lump-type scores; BUS-UCLM's all-image figure is dragged down by normal false positives (320/413).

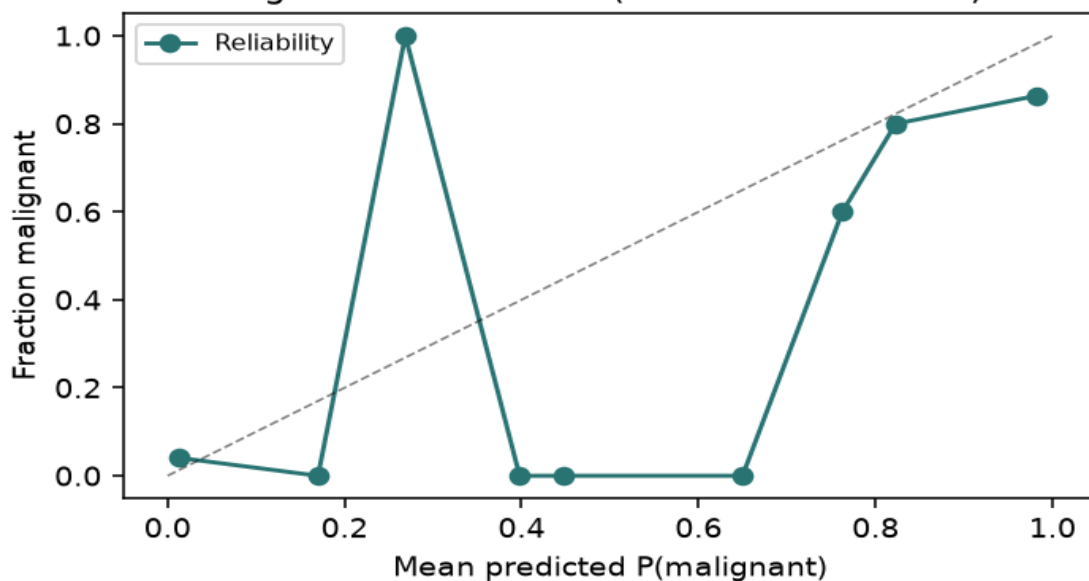
Sources: results/external/busbra.json, breast.json, busuclm.json.

Figure 3. Classification ROC (internal vs external)



Takeaway: the curve that separates benign from malignant looks strong on BUSI and weaker once the hospital and scanner change.

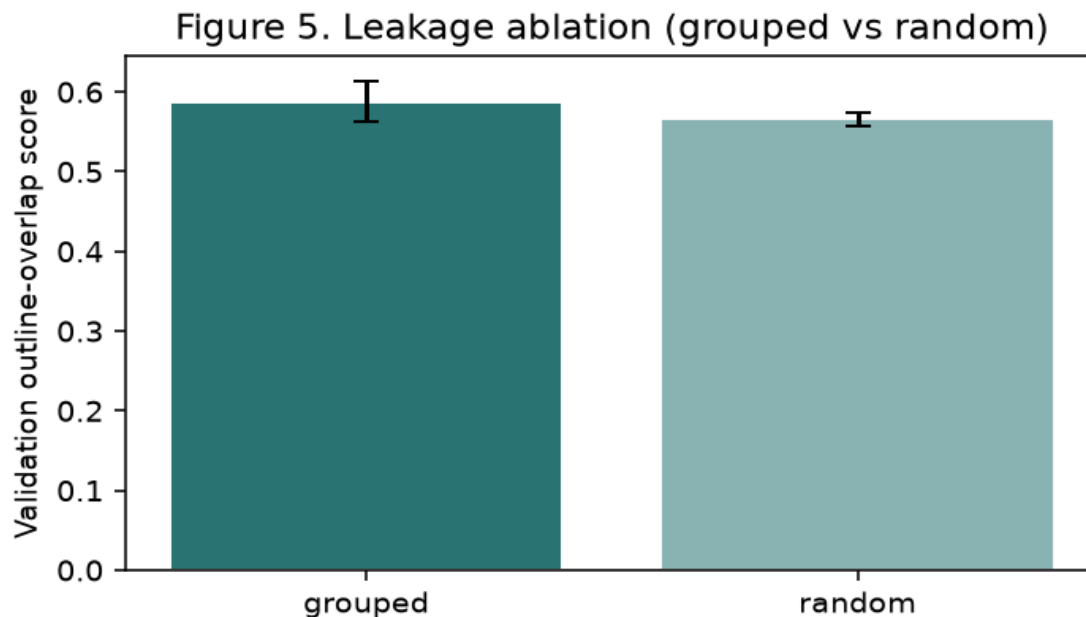
Figure 4. Calibration (BUSI served model)



Takeaway: predicted cancer probabilities are only roughly aligned with observed rates (ECE 0.111) — confidence should be read cautiously.

6.3 Leakage ablation

A short schedule (6 epochs × seeds 42,43,44) trained under grouped versus random splits and evaluated on the same grouped held-out test. Mean validation outline-overlap score did not rise under random splits (Δ random–grouped ≈ -0.022). Residual leakage risk remains because patient identifiers are absent and near-duplicates are documented in the literature.



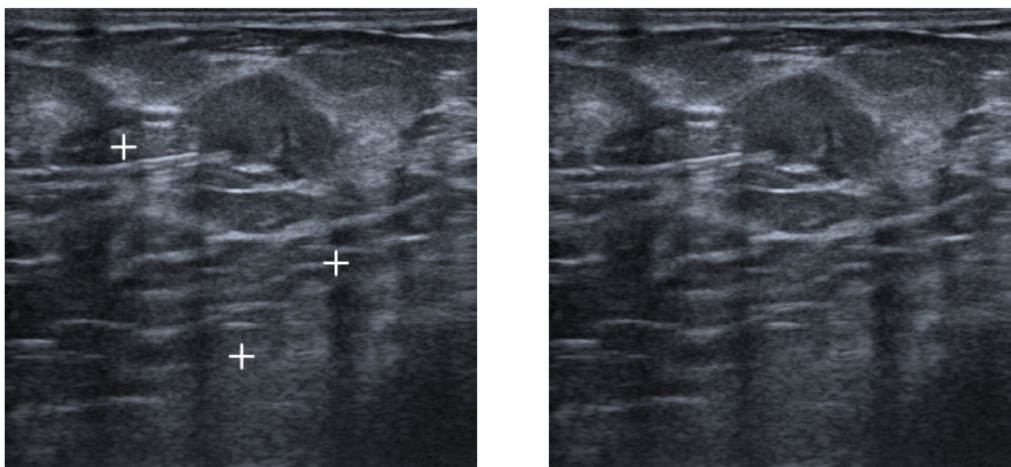
Takeaway: on this short check, random splits did not inflate validation scores versus grouped splits — leakage risk remains for other reasons.

Source: results/leakage_ablation.json.

6.4 Caliper-mark experiments

Images with burned-in measurement marks often look easier: flagged-frame outline-overlap score 0.764 versus clean 0.628. Erasing markers on the original test (experiment E-a) moved overall score from 0.697 to 0.702 and did not collapse the flagged-clean gap. Retraining with inpainting (E-c) produced external scores BUS-BRA 0.626 and BrEaST 0.608 with clean-subset 0.626 and AUC 0.945 — not enough, under the pre-registered swap rule, to replace served v1.0.0.

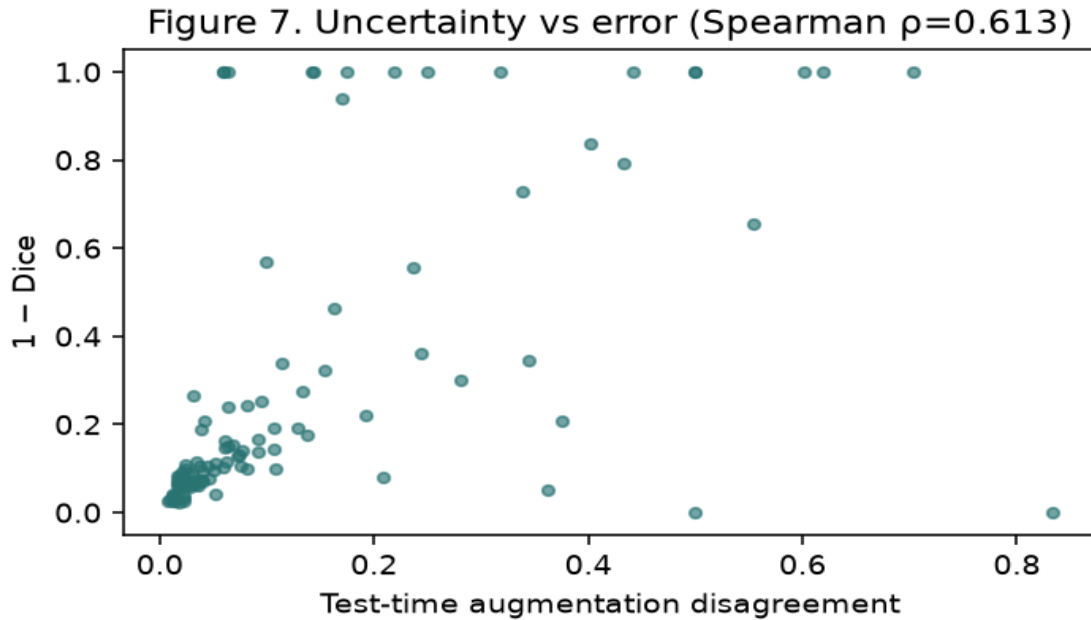
Figure 6. Caliper erase demo (CC BY BrEaST; digitally altered)
Before (synthetic calipers) After Telea inpaint



Takeaway: measurement marks can act as shortcuts; simply erasing them on test images does not automatically make hard clean cases easy.

6.5 Uncertainty analysis

Test-time augmentation (TTA) flips and mild transforms are run offline; disagreement across those views is treated as an uncertainty signal. Spearman correlation between disagreement and 1–Dice is 0.613 on the BUSI test (n=112). Keeping the most certain 80% of cases raises mean outline-overlap score to 0.778. Interpretation: agreement under flips — not a calibrated probability of being wrong.



Takeaway: when the outline stays stable under flips, errors are less common — useful as a research triage signal, not as a clinical confidence meter.

6.6 Measurement agreement (BrEaST)

On BrEaST, model versus expert longest diameter (mm) shows Pearson correlation proxy 0.697 (n=252); area correlation proxy 0.763; disagreement at a 20 mm threshold occurs on about 17.1% of cases. Research only — imaging size is not pathologic T-stage.

6.7 Version-2 retrain and swap-rule outcome

A later candidate trained on BUSI plus patient-grouped BUS-BRA (larger backbone, 256×256, stronger augmentation) was compared with served v1.0.0 under a pre-registered fair swap rule. Seed-mean clean-subset outline-overlap score was 0.614 versus v1 0.623 (Δ -0.009). Same-source BUS-BRA held-out rose to 0.894, and truly external BrEaST to 0.672, with AUC 0.912 and packaged size about 21.3 MB. Because the seed-mean clean-subset rule failed, the site keeps serving v1.0.0 and publishes the comparison as an experiment.

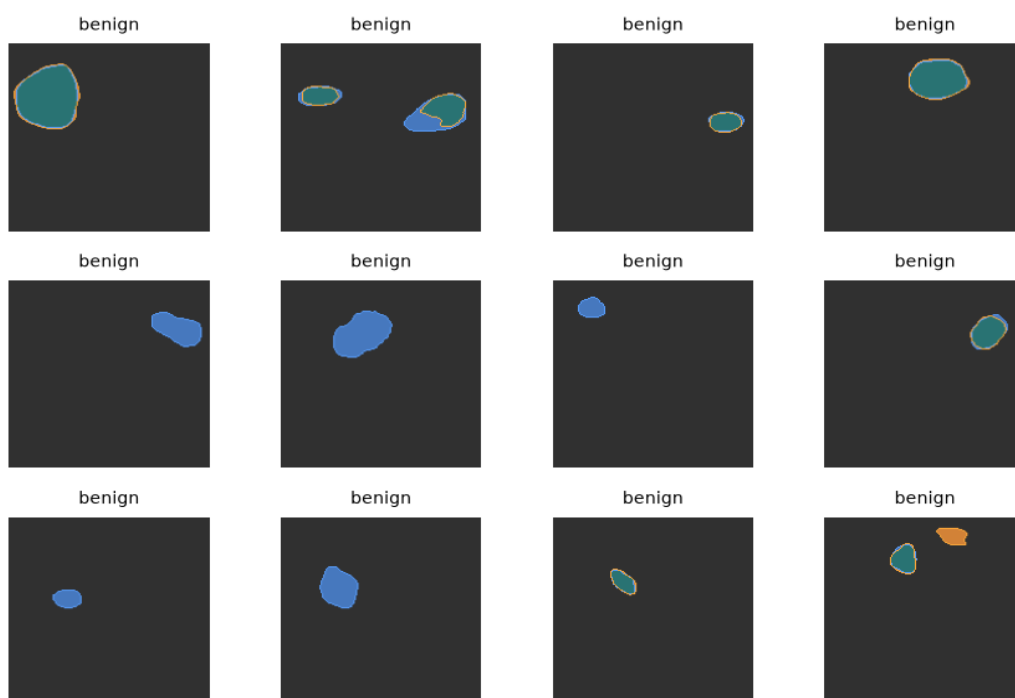
Takeaway: better scores on a mixed-training held-out set are not enough; the published model only changes when every pre-agreed gate passes on the seed mean.

Source: results/v2_experiment.json.

7. Error analysis

The Model Errors explorer on the site lists 105 hard held-out BUSI cases as outline-only silhouettes (BUSI ultrasound pixels withheld because redistribution rights are unclear). Category counts: boundary disagreement 65, false lesion on normal 12, wrong class 11, missed lesion 8, over-segmentation 7, under-segmentation 2. Boundary disagreements dominate; normal false outlines remain the most clinically intuitive failure mode for a triage-style demo.

Figure 8. Failure-case outline silhouettes (BUSI; no ultrasound pixels)



Takeaway: silhouette cards make misses visible without redistributing BUSI pixels — most hard cases are boundary fights, not total collapses.

8. Discussion: what the numbers mean in practice

An outline-overlap score near 0.697 on held-out BUSI means that, on average, the computer outline and the expert outline share most of their area — useful for a demo and for learning what segmentation metrics feel like, not for claiming radiologist parity. External outline scores in the 0.63–0.71 range suggest the spatial task transfers better than the lump-type score, which falls when scanners and labeling habits change. That pattern matches the broader literature on dataset shift.

Normal false positives (12/19 on BUSI; 320/413 on BUS-UCLM) are the practical warning label: a browser demo that “finds” a lump on a healthy-looking frame can mislead a casual reader faster than a slightly jagged outline on a true mass. Calibration (ECE 0.111) is imperfect, so verbal confidence cues are intentionally subdued on the site.

Readers comparing this project with published BUSI leaderboards should weight protocol first. Many papers use random splits on a dataset with documented near-duplicates; this project uses grouped splits, reports the served browser artifact (11.7 MB), and publishes external numbers without retuning. A higher in-domain score under a leakier split is not a stronger scientific claim than a slightly lower score under a stricter one.

The failed v2 swap is part of the result, not an embarrassment to hide. Same-source BUS-BRA held-out rose to about 0.894, and BrEaST to about 0.672, yet the clean-subset gate (v2 mean 0.614 vs v1 0.623) blocked promotion. Pre-registering that rule before seeing the seed mean is how a student project stays honest when a larger model looks tempting on a chart.

For admissions readers and scientifically literate parents, the portfolio claim is process honesty: grouped splits, frozen external tests, a written swap rule that refused a flashy upgrade, and a downloadable write-up whose figures are locked to committed JSON.

9. Limitations

Several constraints bound how far these numbers should travel outside a research demo:

- No patient identifiers on BUSI → residual near-duplicate / patient-leakage risk despite perceptual grouping.
- Domain shift: classification AUC drops from 0.931 internally to as low as 0.638 on BUS-BRA.
- Calipers and on-screen text may still act as shortcuts on annotated frames.
- 160×160 resolution loses fine boundary detail; normal-image false positives remain (12/19).
- Single-author student project without clinical validation; not a medical device.
- A model score is not a BI-RADS category; imaging size is not surgical staging.
- External BUS-UCLM images were scored locally and are not redistributed with the site.

10. Ethics and responsible use

Code is released under the MIT license. The full BUSI archive is not redistributed (redistribution rights are unclear); users must obtain it themselves and cite Al-Dhabyani et al. (2020). External demo imagery uses Creative Commons Attribution collections with credit. The site uses no analytics or cookies; Demo images are processed in the browser only and are not uploaded to a server. This write-up and the live demo are educational research materials. They must not be used for diagnosis, triage, screening, or any care decision. No clinician reviewed this project for clinical deployment, and none is claimed.

11. Future work

Natural next steps, if pursued, include stronger normal-vs-lesion rejection, patient-level collections with clearer licenses, higher-resolution models that still fit a browser budget, and uncertainty displays that a non-specialist can interpret without over-trust. Any promoted weight file would need a new version number and a fresh pass of the written swap rule.

A second thread of future work is communication: shorter silhouette cards that teach error types, clearer captions for parents who are not engineers, and printable one-pagers that keep the same locked numbers as this PDF. None of those changes should quietly retune thresholds on external data.

12. Reproducibility

Software version 1.0.0 is archived with DOI <https://doi.org/10.5281/zenodo.23286597>. To regenerate this PDF: install Python dependencies (reportlab, pypdf, matplotlib), run the report builder script from the repository root, then run the drift check so every locked number still appears in the PDF text. Paper metrics are exported from the same results JSON into LaTeX macros. Served artifact fingerprint (sha256 prefix): 0bbf529d... · packaged size 11.7 MB. Typical software stack for regeneration: Python 3.10+, Node.js 20+ for the web app, onnxruntime / onnxruntime-web for model execution. Exact pinned versions live in the repository lockfiles on the tagged release.

13. AI-use disclosure

Project by Surabhi Fadnavis. The code, analysis, text, and video were produced with AI tools (Cursor).

14. Glossary

Outline-overlap score (Dice)

A number from 0 to 1 describing how much the computer outline and the expert outline overlap. Twice the shared area divided by the sum of both areas; 1 is a perfect match.

Intersection-over-union (IoU)

Shared outline area divided by the area of either outline. Related to Dice; usually a little lower than Dice for the same pair of outlines.

Area under the curve (AUC)

A summary of how well a score ranks cancerous cases above benign ones across all thresholds. 0.5 is chance; 1.0 is perfect ranking.

Sensitivity / Specificity

Sensitivity is the share of truly cancerous cases the score catches at a chosen threshold; specificity is the share of truly benign cases it correctly leaves below that threshold.

Calibration / expected calibration error (ECE)

Whether a predicted probability (for example “70% malignant”) matches how often such cases are actually malignant. ECE summarizes the mismatch across probability bins.

BI-RADS

Breast Imaging Reporting and Data System — a standard lexicon and assessment categories radiologists use. A model score is not a BI-RADS category.

U-Net

A widely used neural-network shape for image outlining: a compressing path, an expanding path, and skip connections that preserve spatial detail.

ONNX

Open Neural Network Exchange — a portable file format for trained models so the same weights can run in Python and in the browser.

Quantization (INT8)

Storing weights with 8-bit integers instead of 32-bit floats to shrink download size and speed inference, with small metric changes when done carefully.

Test-time augmentation (TTA)

Running the model on flipped or lightly altered copies of the same image and measuring how much the outlines disagree — used here as an uncertainty signal.

External validation

Scoring a frozen model on data from other hospitals or collections without retuning, to test whether results were specific to the training source.

Data leakage

Accidental overlap between training and test information (for example near-duplicate frames on both sides) that makes scores look better than true generalization.

15. References

Citations below support the clinical framing, datasets, model family, and software archive. Prefer the dataset papers when quoting collection size or license.

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14. Project repository: <https://github.com/surabhif/Breast-ultrasound-lesion-segmentation>

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